

F24-181-D-Little Learner

Project Team

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Session 2021-2025

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Islamabad, Pakistan**

June, 2025

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Chapter 1

Introduction

This document is a formal proposal, for the development of an interactive AI Academic Tutor for children, our target audience is Pakistani children aged 3-7 years old. This document details the project's objectives, scope, anticipated challenges, and methodologies, serving as a communication tool for stakeholders and a guide for project development teams. It aims to secure approval, funding, and resources while providing a structured execution plan.

Previous Work and Inspiration

Arvind Narayanan's Kids Tutor Idea: This involved a customized voice interface for a young child, which adjusted responses based on the child's comprehension level, showing the potential of AI in early education.[1]

SoapBox Labs: Develops voice recognition technology tailored for children, highlighting the importance of specialized solutions for educational environments. [4]

Speechace API: Offers Voice AI solutions for enhancing K-12 educational experiences, though it lacks training specifically on the regional speech of children from Pakistan/India. [6]

1.1 Problem Statement

Uniform access to quality learning resources and individualized teaching instruction is almost nonexistent in Pakistan's early childhood education system particularly for the three to seven years old. There are mature mismatches in curriculum resource provisions according to the age of the learner, teacher pupil ratio, and lack of competent educators especially in rural areas that have left almost 22 million children within this age group falling short either of formal education or quality education. The use of current technologies where the children of Pakistan are coming from diverse linguistic and cultural

backgrounds negatively influence both learning outcomes and the quality of interactions. This requires that there is a need for a more sophisticated approach to teaching in order to account for regional accents while still creating engaging and personalized lessons to ensure basic language and number skills development. [2]

These concerns is where it is believed that the proposed solution, that is bearing the name ‘Little Learner’, which will act as an AI based academic tutor, will help by providing engaging techniques of learning through advanced strategies of machine learning and speech recognition. This platform would help to narrow this gap and improve preschool education in each socioeconomic stratum of the country’s population by a correspondingly addressing each stratum’s requirements and speech habits. [3]

1.2 Scope

The scope of the project is the development of our app Little Learner, the AI Tutor that is specified for children aged between 3-7 [5]. The Tutor will:

- **Target Audience:** Focus on children aged 3-7, including those with disabilities, ensuring the system is inclusive and accessible to children with various communication challenges.
- **Technological Focus:** Utilize advanced AI technologies, including adapting and fine-tuning and training current technologies to accurately understand and interact with children in their native languages (Urdu). The project will create a custom dataset tailored to the linguistic and accent variations found in children from Pakistan.
- **Educational Goals:** Enhance the educational opportunities for young children by improving foundational language skills through personalized AI-driven interactions. The project aims to make learning more accessible and engaging, thereby fostering greater inclusivity in early childhood education.
- **Model Development:** Fine-tune existing transcription models, followed by development of a prompted intent classification model. This will accurately interpret and respond to children’s voice inputs.
- **Data Collection and Processing:** Creating a custom dataset of Pakistani children(aged 3-7 years old), speaking numbers 0-9. This will include data annotation, pre-processing and structuring of the data.
- **Implementation and Evaluation:** The AI tutor will be refined using this custom dataset, ensuring that the final product is effective for the educational needs of each child.

The project focuses on both technical development of AI models and the practical application of these models.

1.3 Modules

1.3.1 Voice Interaction and Processing Module

This is the core module of the system that will handle the voice-based interaction and processing of the little learner app.

1. **Speech Recognition:** Records and transcribes the children voice using our fine-tuned whisper model, recognising the diverse patterns.
2. **Answer Validation:** Use LLM to properly analyze and understand the children's responses by using the context of the question.
3. **Error Handling:** Sometimes due to random reasons, the system is not able to recognize the response it will ask to speak again.

1.3.2 Content Management and Personalization Module

This module will manage the content based on the kid's preferences and interests. It will also personalize the questions based on the difficulty level of the kid.

1. **Dynamic Content Repository:** Store and manage a library of educational content that will be used to ask the questions.
2. **Personalization:** Customize the questions based on the difficulty level of the kid that is being maintained with time.
3. **Progress Tracking:** Keep a record of the child's progress and then change the difficulty level of questions accordingly.

1.3.3 Data Collection and Processing

This module is responsible for collecting and making the dataset that will be used later to fine-tune our transcription model. Collecting audio through a web page and then processing and labeling those audio files will be done in this module.

1. **Web-Based Audio Collection Interface:** Ensure the interface is intuitive for users, potentially children or their guardians, to submit audio samples.

2. **Audio Processing** : Apply audio processing techniques to clean up the recordings, such as noise reduction and volume normalization.
3. **Data Labeling** : Manually Label all the audio files.

1.3.4 Security and Scalability Infrastructure Module

Security is critical, especially when dealing with children's data. This module ensures that the platform is secure and scalable to handle varying loads and expand to new regions without degradation in performance.

1. **Data Encryption and Security**: Implement robust encryption methods to secure children's data both at rest and in transit.
2. **Scalable Architecture**: Design the backend to scale resources dynamically based on the demand, ensuring seamless performance during peak usage.
3. **Analytics and Reporting**: Provide tools for educators and parents to view detailed reports on children's progress and system usage.

1.3.5 Web Application Development

Creating and maintaining an interactive and attractive application for the kids to enjoy themselves while using this application.

1. **UI/UX**: Beautiful design and flow to keep kids interested in learning through the application.
2. **Development**: Complete design implementation according the UI/UX.
3. **Integration**: Integrate the model to work with the web application.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Children (End Users)	The "Little Learner" AI tutor is introduced with children of aged range of 3-7 years aiming at providing effective, interactive and fun learning content in accordance with individual's level of understanding. This system includes the method of recognizing and understanding children's speech for the preparation of educational content with the use of efficient voices that would help enhance a child's language, numbers, and mind in personalized ways.
Parents / Guardians	The operatives of the 'Little Learner' system must include parents and guardians of the child users. Their main concern would be to follow their child's educational progress and effectiveness of the content used. Parents are familiar with the system and request security with data, settings regulations, progress reviewing, assistance provision, and overall productive and safe learning for children.
System Ad- ministrators	The system administrators are the ones who ensure that the "Little Learner" system functions correctly and is used safely. This function includes governance of the technical stack, maintenance of system availability, organization of information security protection, and activity regulation considering educational and information privacy aspects. They are in charge of performing routine maintenance and providing technical support to the system whenever it becomes necessary.

User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

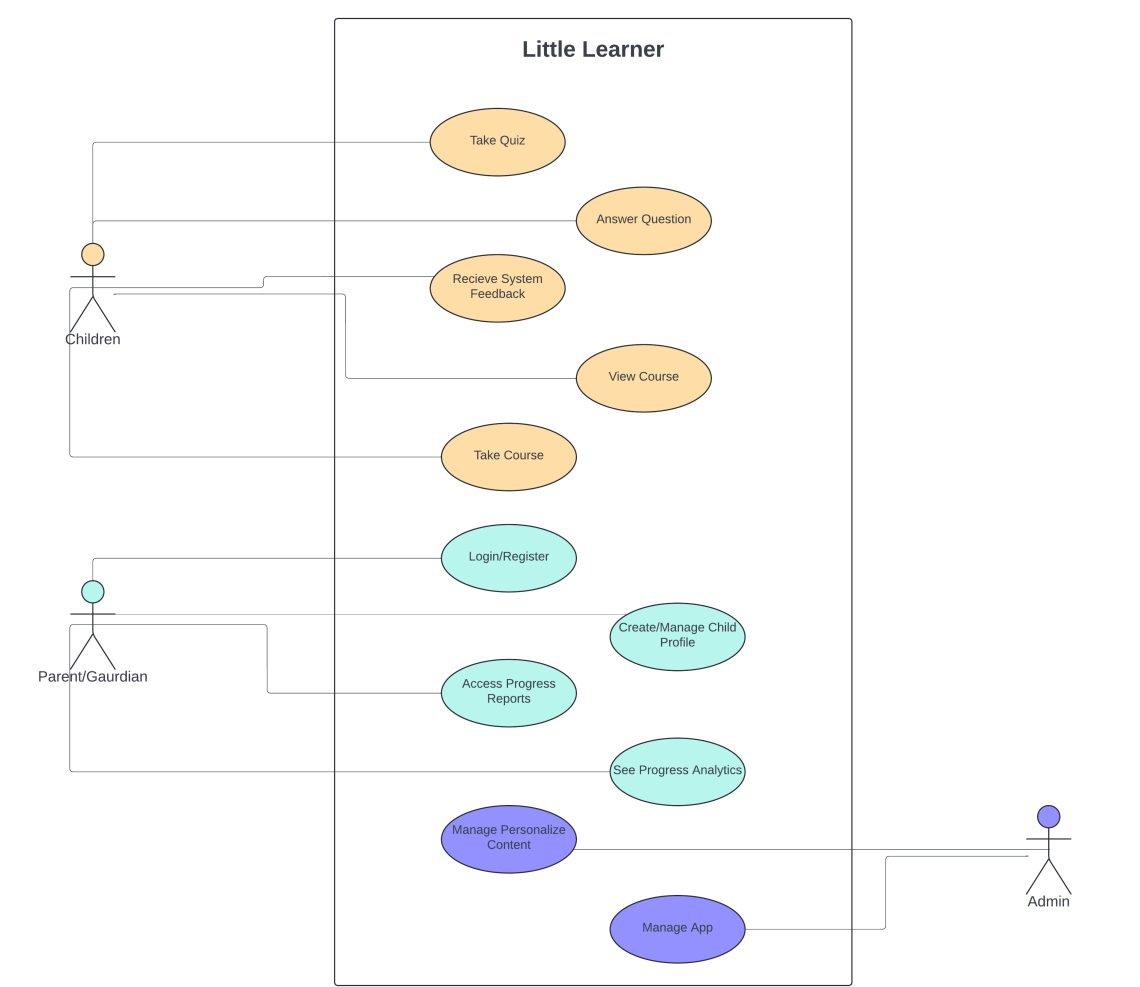


Figure 2.1: Use Case Diagram

2.1.1 Fully Dressed Use Cases

2.1.1.1 Use Case: Take Quiz

Primary Actor: Child

Secondary Actor: Parent/Guardian

Preconditions:

- The parent/guardian is registered and logged into the system.
- The parent/guardian has selected the child's profile.
- The child has selected a course that includes quizzes.

Postconditions:

- The quiz results are recorded in the system.
- The child receives immediate feedback on their performance.
- The child's progress is updated based on the quiz results.

Main Success Scenario:

1. The parent/guardian logs into the app.
2. The parent/guardian selects the child's profile from multiple profiles (if applicable).
3. The child selects a specific course to engage with.
4. The child navigates to the quiz section of the selected course.
5. The child selects a quiz to take.
6. The system presents the quiz questions one by one.
7. The child submits answers to the quiz questions.
8. The system evaluates the answers in real-time.
9. The system provides immediate feedback on each question after submission.
10. Upon completing all questions, the system displays the quiz results and updates the child's progress.

Alternative Flows:

A1. Multiple Attempts Allowed:

- If the quiz policy allows multiple attempts, the child can retake the quiz.
- Steps 6 through 10 of the Main Success Scenario are repeated.

Exception Cases:

E1. Incorrect Login Credentials:

- If the parent/guardian enters incorrect credentials, the system displays an error message.
- The parent/guardian is prompted to try logging in again or to reset the password.

E2. Profile Selection Error:

- If there's an error in selecting the child's profile (e.g., system error, profile not found), the system notifies the parent/guardian.
- The parent/guardian is prompted to try again or contact support.

E3. Course/Quiz Access Error:

- If the selected course/quiz is unavailable or locked, the system informs the child and provides reasons (e.g., prerequisite not met, course/quiz under maintenance).
- The child (or parent/guardian) is advised to select another course or wait until access is restored.

E4. No Response During Quiz:

- If the system detects no input from the child within a specific time, it prompts the child to respond or offers to pause/exit the quiz.
- If no action is taken despite reminders, the quiz session is paused or ended based on the system settings.

E5. Technical Issues (e.g., loss of connectivity):

- If there is a loss of connectivity during the quiz, the system attempts to save the last recorded state.
- The child is prompted to reconnect. Upon reconnection, the child can resume from the saved state or restart the quiz.

2.1.1.2 Use Case: Answer Question

Primary Actor: Child

Preconditions:

- The child is logged into their profile.
- The child is actively engaged in a course or a quiz.

Postconditions:

- The child's answer is recorded in the system.
- Feedback is provided based on the correctness of the answer.
- The child's progress and performance data are updated.

Main Success Scenario:

1. The child is presented with a question, either as part of a quiz or interactive learning session.
2. The system waits for the child's response.
3. The child submits an answer using their voice.
4. The system uses speech recognition to transcribe the child's answer.
5. The system evaluates the answer based on the question's context and expected response.
6. The system provides immediate feedback to the child:
 - If the answer is correct, the system praises the child or encourages further interaction.
 - If the answer is incorrect, the system offers hints or explains the correct answer.
7. The child's response and the feedback are recorded in their learning progress.

Alternative Flows:

A1. Request for Repeat:

- If the child does not understand the question, they can request it to be repeated.

- The system repeats the question without penalty.

Exception Cases:

E1. Speech Recognition Error:

- If the speech recognition system fails to accurately transcribe the child's response, the system prompts the child to repeat their answer.
- If repeated attempts fail, the system may suggest the child to type their answer if possible.

E2. No Response Detected:

- If the system detects no response within a reasonable timeframe, it prompts the child to answer.
- If there is still no response, the system may skip the question or end the session based on the settings.

E3. Incorrect Response Format:

- If the child submits an answer in an incorrect format (e.g., typing in a voice-only session), the system notifies the child of the proper format and requests a new answer.

E4. System Error/Technical Issue:

- If a system error occurs during the question-answer session, the system attempts to log the issue and may restart the session.
- The child is informed about the technical issue and is asked to retry or resume the session later.

2.1.1.3 Use Case: Receive System Feedback

Primary Actor: Child

Preconditions:

- The child has submitted an answer to a question or completed a quiz.
- The system has evaluated the child's answer or overall quiz performance.

Postconditions:

- The child receives feedback that is informative and tailored to their response.
- The feedback contributes to the child's learning progress and understanding.

Main Success Scenario:

1. After the child submits an answer, the system processes and evaluates the response.
2. The system determines the correctness of the answer:
 - If correct, the system calculates and displays points earned and may provide additional information or praise.
 - If incorrect, the system explains why the answer was wrong and provides the correct answer or a hint for better understanding.
3. The feedback is presented in a child-friendly manner, using positive reinforcement to encourage learning.
4. If the feedback involves corrective information, it includes educational content to help the child understand the correct concept.
5. The child listens to or views the feedback.
6. The feedback details (content, timing, and child's response) are recorded for future learning analytics and adaptive learning improvements.

Alternative Flows:**A1. Continuous Feedback During Quizzes:**

- Instead of waiting until the end of a quiz, the system provides immediate feedback after each question.
- This immediate feedback helps correct misconceptions early and maintains engagement.

A2. Feedback Through Different Media:

- Depending on the child's learning preference and the complexity of the topic, feedback is delivered via text, audio, or visual aids (like charts or animations).

Exception Cases:**E1. Feedback Misunderstanding:**

- If the child indicates or the system detects that the feedback was not understood (e.g., the child repeats the mistake), the system offers to rephrase or provide additional examples.
- The system may invite the child to ask questions if they need more explanation.

E2. Technical Issues Affecting Feedback Delivery:

- If there is a delay or failure in delivering feedback due to technical issues, the system informs the child of the issue and attempts to resolve it.
- If the issue persists, the system saves the session state and advises the child to restart the application or come back later.

E3. Inaccurate Feedback:

- If erroneous feedback is given due to a system error in content or assessment logic, the system logs the error for review.
- The child is advised that there was an error, and the correct feedback is provided after a brief correction period.

2.1.1.4 Use Case: View Course / Take Course

Primary Actor: Child

Preconditions:

- The child is logged into their profile.
- The child has navigated to the course section of the app.

Postconditions:

- The child has accessed and interacted with the course content.
- The system has recorded the child's interactions and progress for future reference and adaptive learning adjustments.

Main Success Scenario:

1. The child selects a course from a list of available courses.
2. The system displays the course details, including a brief overview, learning objectives, and any prerequisites.

3. The child starts the course, which begins with an introductory video.
4. Throughout the video, there are interactive pauses where the video stops, and the system asks the child questions related to the content just covered or about to be covered.
5. The child responds to these interactive questions using voice inputs.
6. The system processes the child's responses and provides immediate feedback:
 - If the response is correct, the video continues with positive reinforcement.
 - If the response is incorrect, the system provides the correct information and a brief explanation before continuing.
7. The course includes several sections, each with its own set of videos and interactive pauses.
8. The child's progress, including their answers and the system's feedback, is recorded and analyzed for adaptive learning purposes.

Alternative Flows:**A1. Course Resumption:**

- If the child exits the course midway, the system saves their progress.
- When the child returns, they are given the option to resume where they left off or restart the course.

A2. Additional Learning Materials:

- After answering questions or at the end of a section, the child can access additional learning materials such as worksheets or extended reading.

Exception Cases:**E1. Response Misinterpretation:**

- If the system misinterprets the child's voice response due to background noise or unclear articulation, it prompts the child to repeat their answer.
- If issues persist, the system may offer the child the option to input their response through an alternative method, such as selecting from multiple choices.

E2. Technical Issues with Video Playback:

- If there are issues with video playback (e.g., buffering, errors), the system attempts to resolve the issue and notifies the child of the delay.
- If the issue cannot be resolved promptly, the child is advised to try accessing the course later, and the incident is logged for technical review.

E3. Lack of Engagement:

- If the system detects a lack of engagement (e.g., no responses to interactive questions, multiple incorrect answers suggesting distraction), it prompts the child with motivational messages or hints to re-engage them.
- The system may also notify parents or educators if it observes consistent disengagement, suggesting further intervention.

2.1.1.5 Use Case: Login/Register

Primary Actor: Parent/Guardian

Preconditions:

- The parent/guardian has access to the device with the app installed.
- The app is functioning properly without technical issues.

Postconditions:

- The parent/guardian is logged into the system, gaining access to manage their child's profile and view progress.
- If registering, the parent/guardian has created a new account with credentials stored securely.

Main Success Scenario:

1. The parent/guardian opens the app and is presented with options to either log in or register.
2. **For Login:**
 - (a) The parent/guardian enters their username and password.
 - (b) The system verifies the credentials and logs them into the system.
 - (c) The parent/guardian gains access to their dashboard where they can manage profiles, view progress, and adjust settings.

3. For Registration:

- (a) The parent/guardian selects the register option.
- (b) They are prompted to provide necessary information, including name, email address, password, and contact details.
- (c) They may also be asked to set up security questions or enable two-factor authentication for enhanced security.
- (d) After submitting the information, the system creates the account and sends a confirmation email to the provided email address.
- (e) The parent/guardian confirms their email address by clicking a link in the email, which activates their account.
- (f) Upon successful registration and email verification, the parent/guardian is redirected to log in with their new credentials.

Alternative Flows:**A1. Forgotten Password:**

- If a parent/guardian forgets their password, they select the “Forgot Password” option.
- They enter their email address, and the system sends them an email with a link to reset their password.
- They follow the link, reset their password, and are then able to log in with the new credentials.

A2. Update Profile Information:

- After logging in, the parent/guardian can access their account settings to update profile information or change their password.

Exception Cases:**E1. Incorrect Credentials:**

- If the entered credentials are incorrect, the system displays an error message.
- The parent/guardian is given the option to try logging in again or to reset their password.

E2. Registration Errors:

- If there is an issue during registration (e.g., missing information, weak password), the system provides specific feedback about what needs to be corrected.
- The parent/guardian corrects the information and resubmits the form.

E3. Email Verification Issues:

- If the parent/guardian does not receive the confirmation email, they can request the email be resent.
- If there is still an issue, they are advised to check their spam folder or contact support for further assistance.

E4. Technical Problems:

- If there are technical issues preventing login or registration (e.g., server down, connection error), the system notifies the parent/guardian of the issue and suggests trying again later.

2.1.1.6 Use Case: Create/Manage Child Profile

Primary Actor: Parent/Guardian

Preconditions:

1. The parent/guardian is registered, logged in, and has verified their account.
2. The parent/guardian has access to the dashboard or profile management section of the app.

Postconditions:

1. A new child profile is created, or existing profile information is updated successfully.
2. Changes are stored securely and are reflected immediately in the app for future sessions.

Main Success Scenario:

1. The parent/guardian navigates to the “My Children” section of the app after logging in.
2. **For Creating a New Profile:**

- (a) The parent/guardian clicks on “Add Child” or similar button.
- (b) They are prompted to enter details about the child, such as name, age, educational level, interests, and any specific learning preferences or needs.
- (c) They may also upload a photo or choose an avatar for the child’s profile.
- (d) Upon entering all required information, the parent/guardian submits the form.
- (e) The system processes and confirms the creation of the new child profile, providing a unique ID for the child’s profile within the app.

3. For Managing an Existing Profile:

- (a) The parent/guardian selects an existing child’s profile from a list.
- (b) They click on “Edit Profile” or a similar option.
- (c) They update any details as needed, such as changing the child’s educational level or updating interests.
- (d) They can also deactivate or reactivate the profile, or update privacy settings.
- (e) After making the desired changes, they submit the updates.
- (f) The system processes and confirms the updates to the child’s profile.

Alternative Flows:

A1. Removing a Profile: If a parent/guardian decides to remove a child’s profile, they select the delete option within the child’s profile settings. The system prompts for confirmation to prevent accidental deletions. Upon confirmation, the profile is deleted, and the system updates the list of profiles accordingly.

Exception Cases:

E1. Incomplete Profile Information: If essential information is missing when attempting to create or update a profile, the system displays an error message specifying the missing data. The parent/guardian is prompted to complete these fields before proceeding.

E2. Privacy Concerns: If the parent/guardian sets privacy settings that restrict certain app functionalities, they are informed of these restrictions and their implications. They must confirm these settings to proceed with their choices.

E3. Technical Issues: If there are technical issues (e.g., server errors, connectivity issues) that prevent profile creation or updates, the system notifies the parent/guardian and logs the issue. They are advised to try again later, and support may be contacted if the problem persists.

2.1.1.7 Use Case: Access Progress Reports/Analytics

Primary Actor: Parent/Guardian

Preconditions:

1. The parent/guardian is logged into the system.
2. The parent/guardian has access to the child's profile.

Postconditions:

1. The parent/guardian views the child's educational progress in a detailed report.
2. The system logs the access for audit and future reference.

Main Success Scenario:

1. The parent/guardian navigates to the "Progress Reports" section after selecting their child's profile.
2. The system displays a list of available reports, categorized by date, subject, or other relevant criteria.
3. The parent/guardian selects a report to view.
4. The system presents a detailed progress report, which may include:
 - Scores and outcomes of quizzes and assessments.
 - Time spent on various courses and activities.
 - Progress over time in graphical formats.
5. The parent/guardian can download the report as a PDF or print it directly from the system.

Alternative Flows:

A1. Custom Report Generation: The parent/guardian chooses to create a custom report by selecting specific criteria and date ranges. The system generates the report based on the selected parameters.

Exception Cases:

- E1. No Reports Available:** If no reports are available (e.g., if it's the beginning of a course or insufficient data), the system notifies the parent/guardian accordingly. The parent/guardian is advised on when to expect the first report based on the child's activities.
- E2. Technical Issues Accessing Reports:** If there are technical difficulties (e.g., server error, loading failure), the system displays an error message. The parent/guardian is advised to retry accessing the reports later and is given the option to contact support if the issue persists.
- E3. Incorrect Report Details:** If the report contains errors or outdated information, the parent/guardian can report the issue through a feedback option. The system logs the issue and notifies the administration to review and rectify the error.
- E4. Privacy or Security Concerns:** If there are concerns about data privacy or security breaches, the parent/guardian can alert the system through a secure channel. The system investigates and ensures that all data complies with privacy laws and security standards.

2.2 Event Response Table

The next table summarises system responses that are designed based on specific voice interaction events that occur within the voice-assisted AI tutor application. For each entry, the event that triggers the system response, the current state of the system when the event occurs, and the resultant actions by the system are noted down accordingly.

Event	System State	System Response
Child speaks answer	Waiting for answer	Record audio, transcribe speech, validate answer, provide feedback based on correctness.
Child requests question repeat	Active question session	Repeat the question without any penalty or changes to state.
Child gives unclear response	Answer received but not understood	Ask the child to repeat the answer or provide clarification options.
No response from child	Awaiting response	Prompt the child again; if no response after set time, log inactivity and potentially end session.
Background noise detected	Recording answer	Filter out noise, ask child to repeat if necessary, enhance voice recognition accuracy.
Child initiates new interaction	Any state	If appropriate, interrupt current process and address new request or query from the child.
System detects voice command error	Processing command	Log the error, provide error message to child, and offer to restart or continue.
Feedback requested by child	Feedback session	Provide detailed explanation or additional information about the last topic or answer.
Error in voice recognition software	Processing input	Log technical issue, notify child of the error, and provide alternative interaction method if persistent.

Table 2.1: System Response Table for Voice Interactions

2.3 Functional Requirements

2.3.1 Voice Interaction and Processing Module

1. **Audio Input Capture:** The system should be able to capture the kid's voice clearly for processing.
2. **Noise Reduction:** The system should be able to remove noise from the audio input to give better feedback.
3. **Feedback Generation:** The system should be able to produce feedback according to the processed voice. If the answer is correct, then it should encourage the kid. If the answer is wrong, it should provide hints.
4. **Response Re-prompting:** The system should re-prompt the question again if the kid fails to answer correctly or answer clearly after some attempts.
5. **Logging Errors:** The system should store logs of errors so that it can learn the kid's weak points and improve them.

2.3.2 Content Management and Personalization Module

1. **Content Storage:** The system should be able to store content for kids.
2. **Content Updates:** The system should be flexible to add new content or change previous content.
3. **Content Levels:** The system should contain contents for each kid's difficulty level. It should contain content for easy, medium, and hard levels.
4. **Progress Reports:** The system should be able to display full report on the kid's performance like what is the kid's weakness?

2.3.3 Data Collection and Processing

1. **Recording Functionality:** The interface should give users the functionality to record audio files from any device with a microphone.
2. **Noise Reduction:** The collected audio files should be processed in order to remove any unnecessary audio.
3. **Volume Normalization:** The audio files should have equal volume intensity for better training.

4. **Quality Check:** The audio files should be checked manually if the kid is even saying the right words or not.

2.3.4 Security and Scalability Infrastructure Module

1. **Progress Tracking Dashboard:** The parents can view their child's progress like weaknesses and strengths.
2. **Cloud Infrastructure:** The system infrastructure will be run on cloud to ensure scalability.
3. **Secure Authentication:** The system will ensure password protected entry.
4. **Data Encryption:** The data stored or transmitted should be encrypted in order to improve security.

2.3.5 Web Application Development Module

1. **Child-Friendly Design:** The system should be very simple and intuitive to attract the child.
2. **Responsive Design:** The system should be able to work on any device for more accessibility.

2.4 Non-Functional Requirements

2.4.1 Reliability

Making the system accessible to everyone at everytime is very important. The system should be running 24/7 to ensure access at any time. This will ensure that the app is available to users at any time and will boost productivity.

2.4.2 Usability

USE-1: The application will be kept simple and ensures that users can access every section within 2-3 clicks.

USE-2: The application will use visual cues and animations to guide users through sections and learning.

USE-3: The application will be able to provide feedback within 2-3 seconds after user

interaction.

2.4.3 Performance

PER-1: The application will ensure that about 90 percent of features respond in 3-5 seconds.

PER-2: The system will be able to handle 5000 concurrent users at a time without any delay.

PER-3: Audio processing requests will take almost 5-30 seconds to respond varying on the audio quality.

PER-4: The system will ensure that 95 percent of learning content loads within 2-5 seconds of user interaction, if the internet connection is fair.

2.4.4 Security

SEC-1: The system will be able to prevent unauthorized access to user data by encrypting it. To decrypt this information, it will be nearly impossible and brute-force attack will take years.

SEC-2: The system will log all attempts of entry and set restrictions based on these attempts. e.g if a person enters their password incorrect several times, he will be forced to do 2 step-verification.

SEC-3: The system will be able to protect against common security threats like SQL Injection etc. The success rate to stop these attacks will be about 90 percent.

2.5 Domain Model

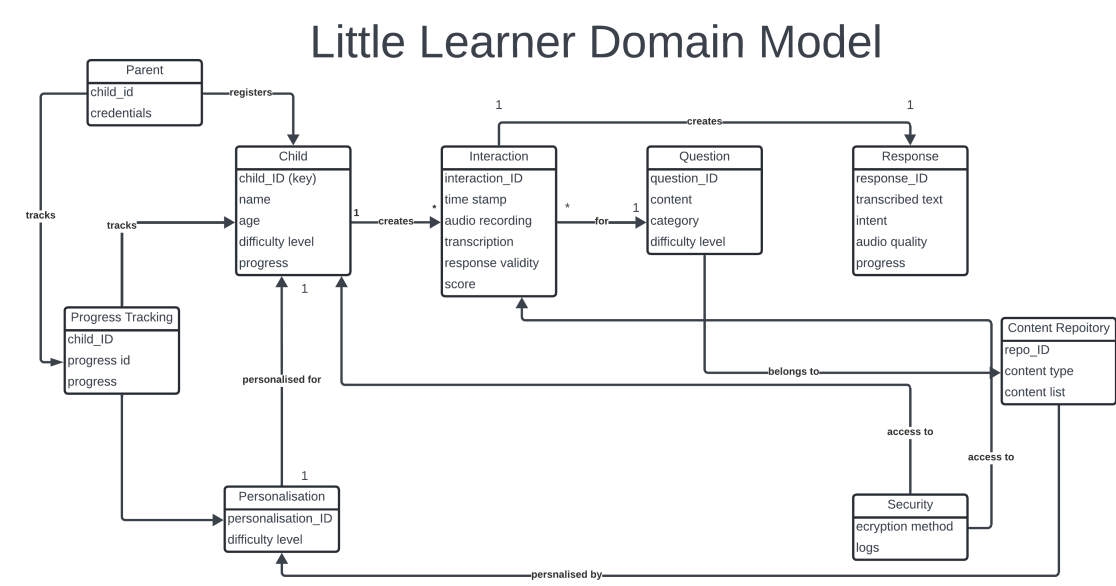


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

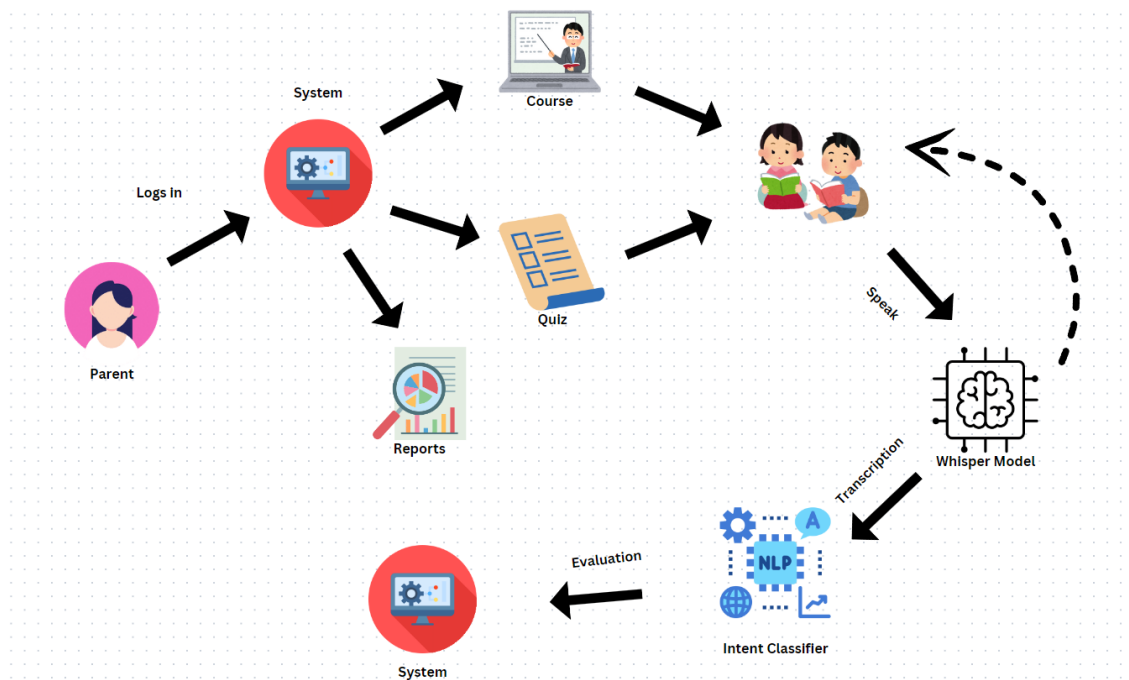


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 State Transition Diagram

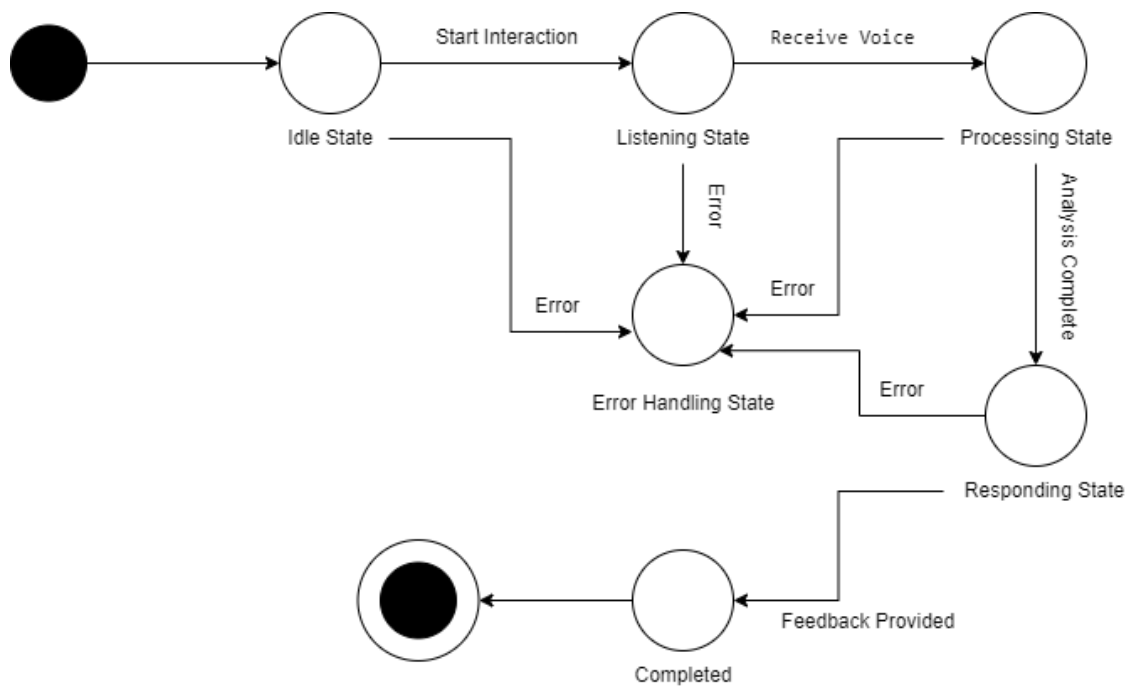


Figure 3.2: State Transition Diagram

3.2.2 Activity Diagram

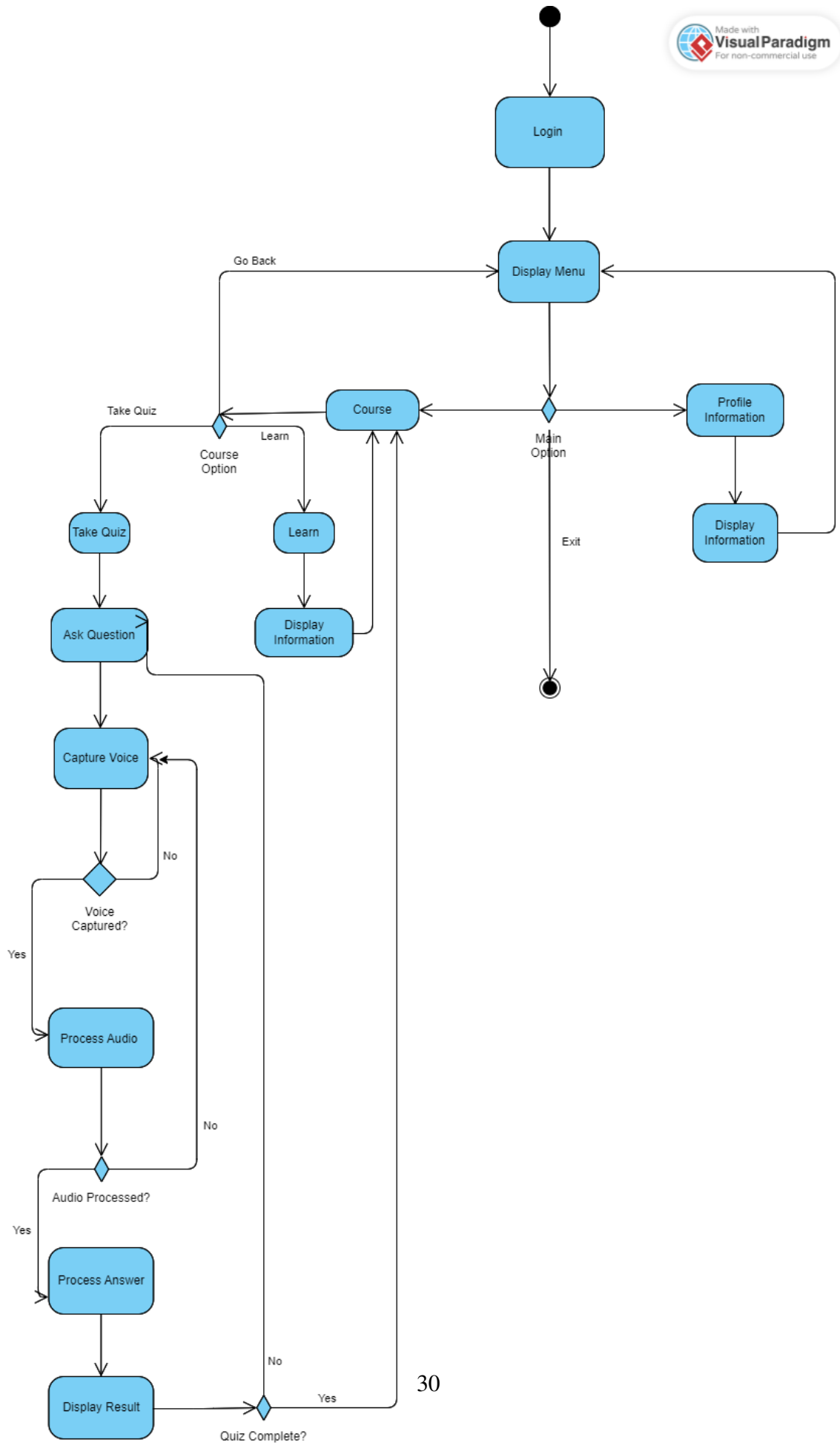


Figure 3.3: Activity Diagram

3.2.3 Parent Registers Child (User Registration)

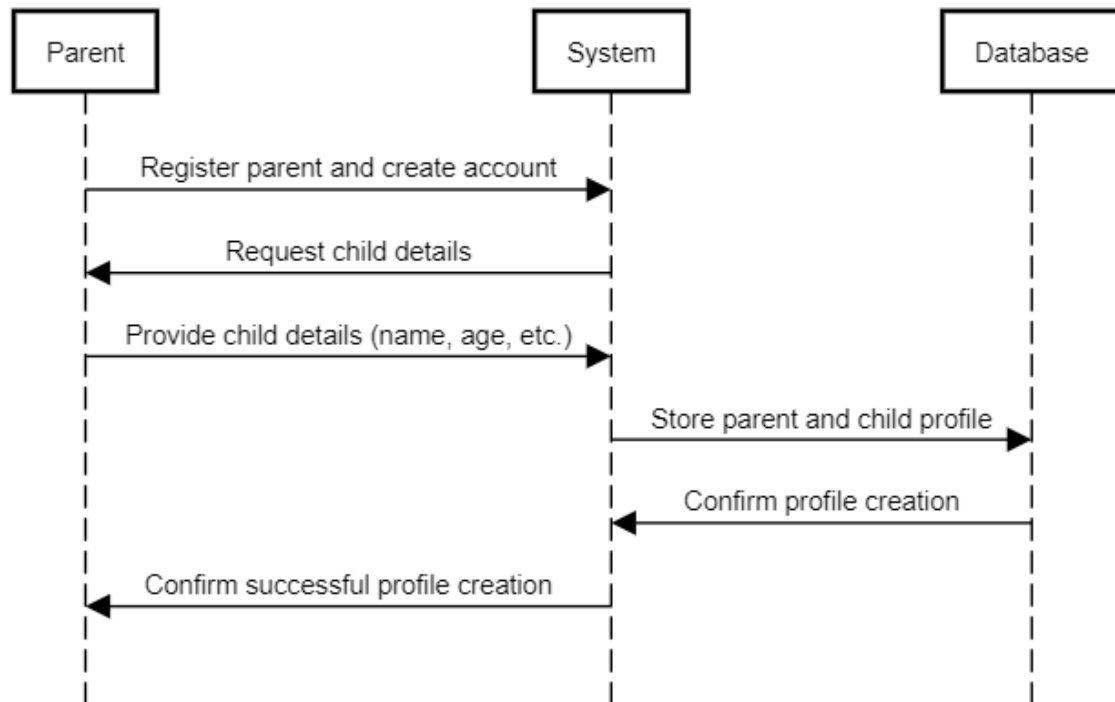


Figure 3.4: Parent Registers Child SD

3.2.4 Child Logs In and Views Progress

3.2.5 Child Learning from Course (Interactive Learning with Questions)

3.2.6 Parent Viewing Child Progress

3.2.7 Child Taking Quiz

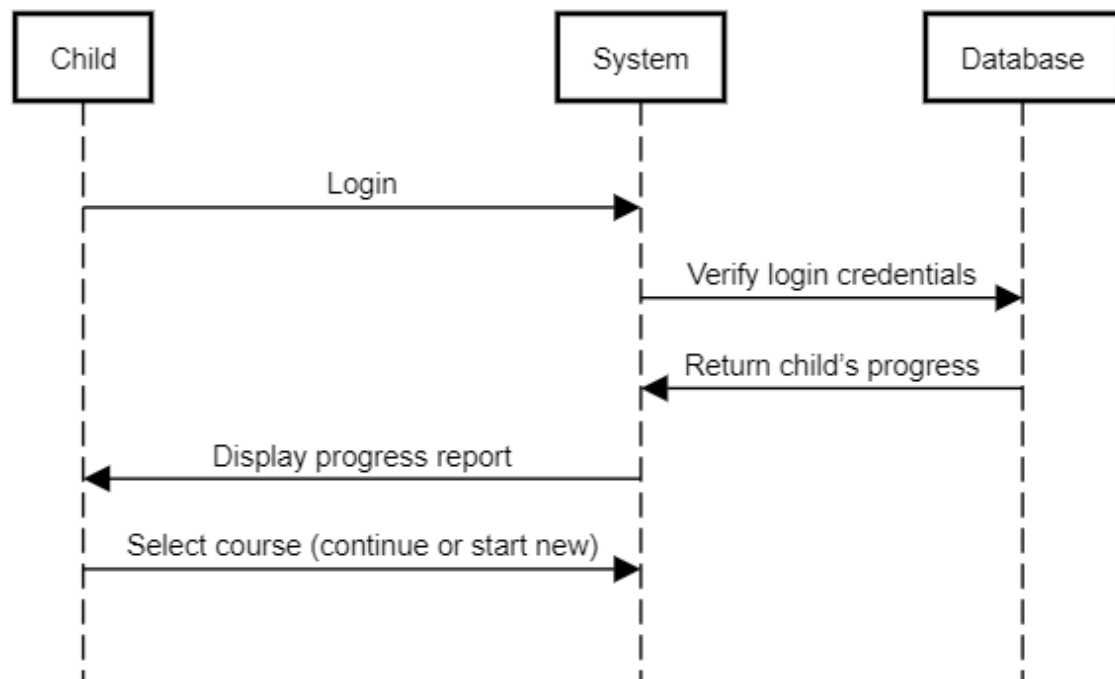


Figure 3.5: Child Logs In and Views Progress SD

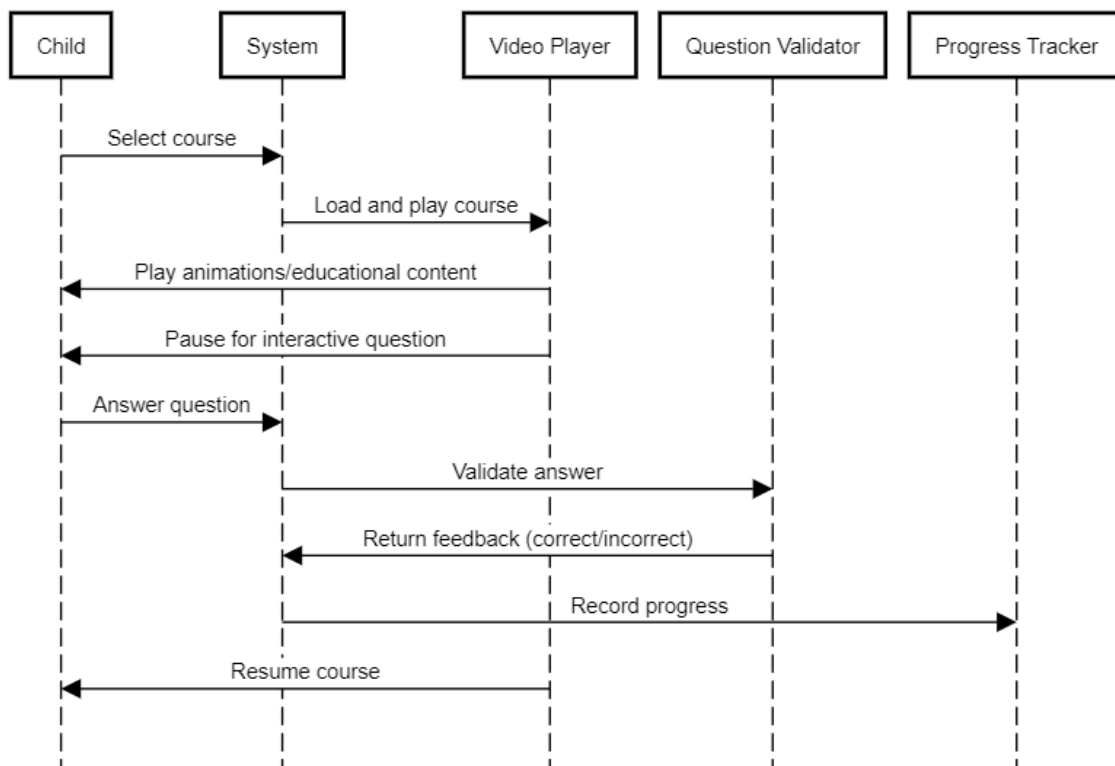


Figure 3.6: Child Learning from Course SD

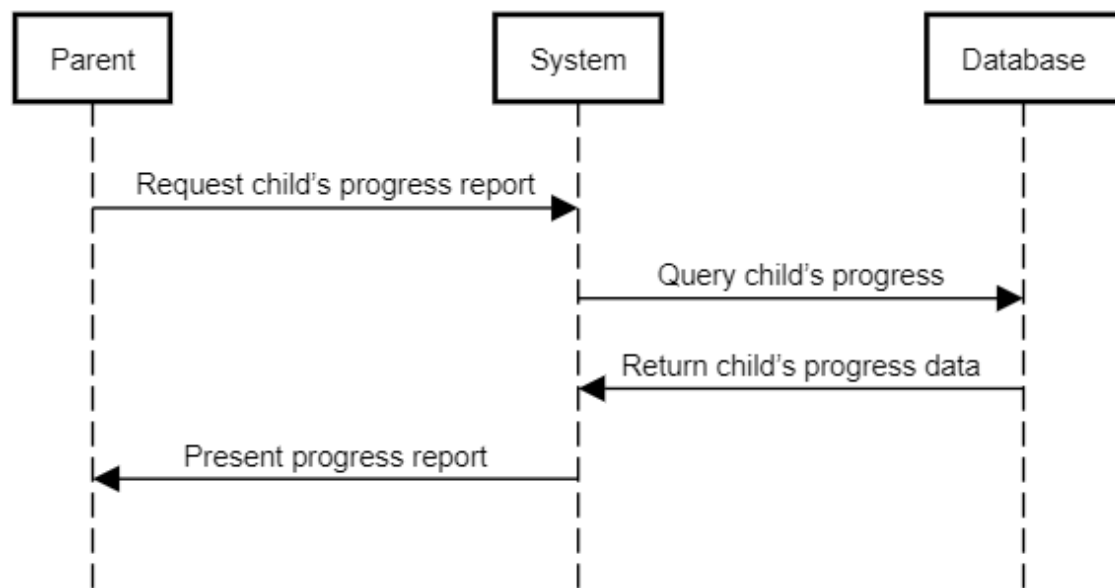


Figure 3.7: Parent Viewing Child Progress SD

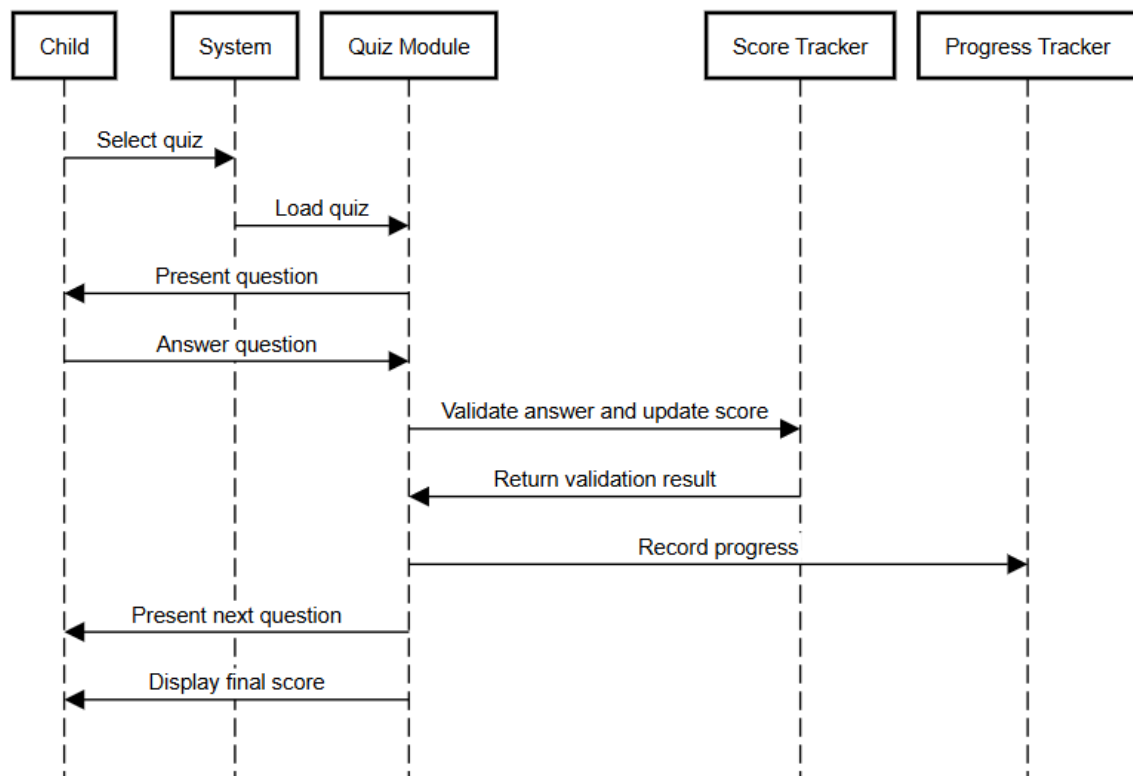


Figure 3.8: Child Taking Quiz SD

3.2.8 Class Diagram

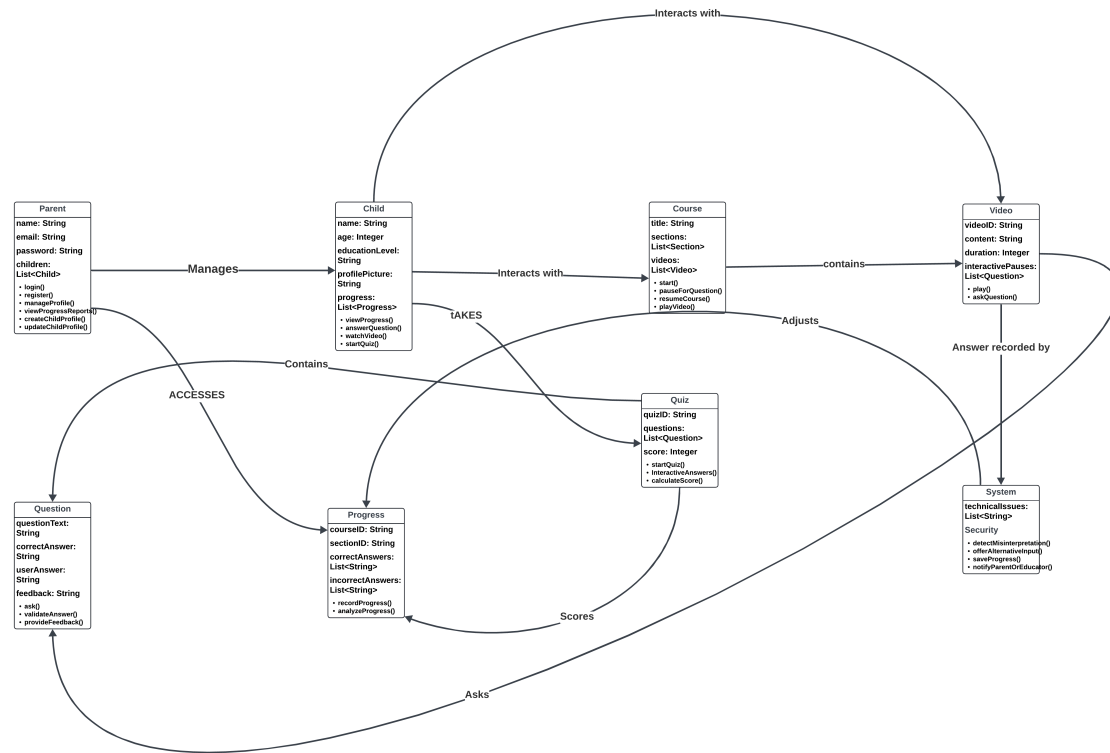


Figure 3.9: Class Diagram

3.3 Data Design

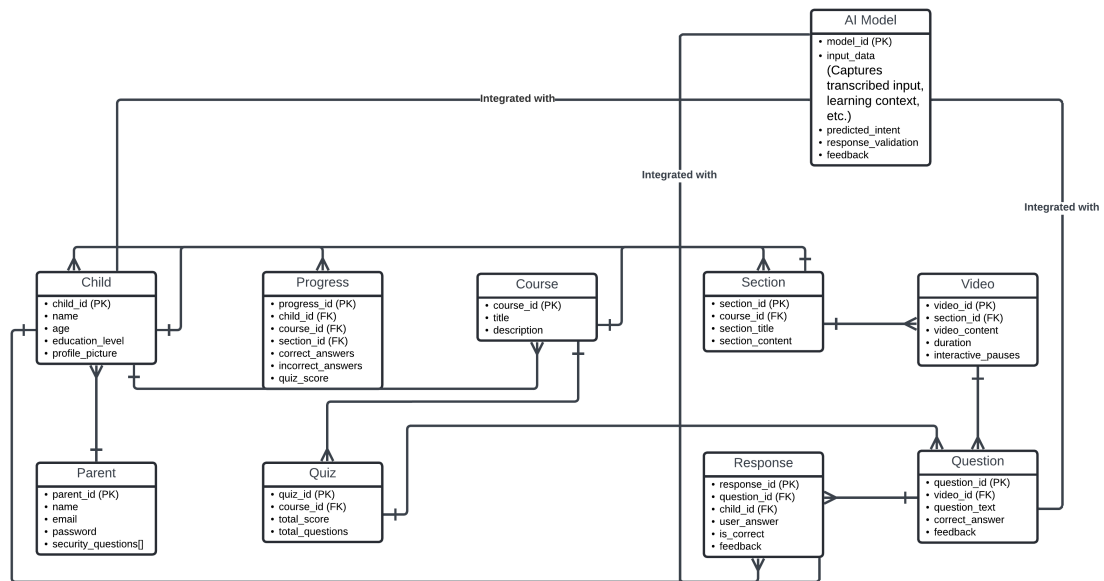


Figure 3.10: Data Design

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